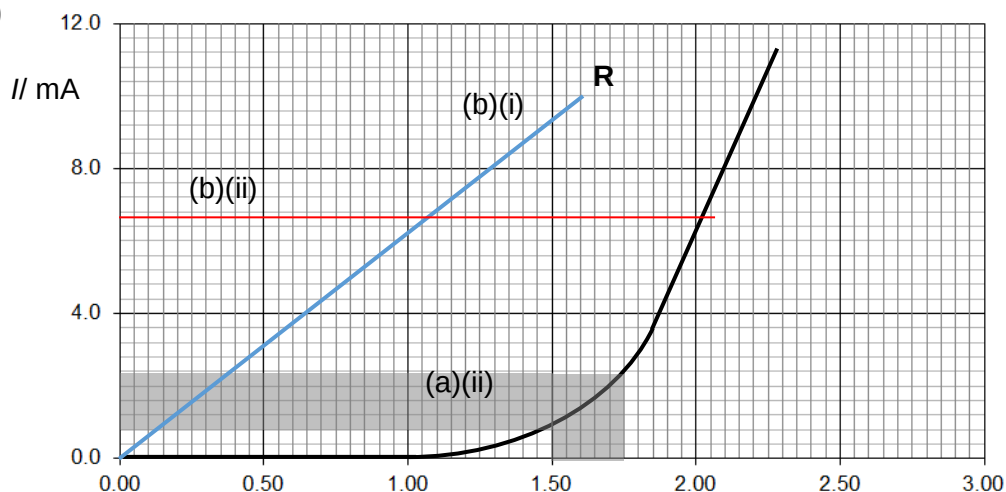


4 (a) (i) $2.25/0.0108 = 208 \Omega$

C1

(a)(ii) & (b) (i)



V/ V

(a)(ii) correct area shaded

B1

(b)(i) Line R should be a straight line passing through (0,0) and (1.60 V, 10.0 mA)

M1

(b) (ii) Draw a horizontal line on Fig. 4.1 such that the pd across 160Ω and LED1 adds up to 3.0 V.

B1

$I = 6.2 \text{ mA}$

A1

(iii) With two LEDs, effective resistance of the circuit is lowered, leading to higher current from battery.

B1

B1

With higher current, the potential difference across the 160Ω resistor is larger, leaving less potential difference across the LEDs.

B1

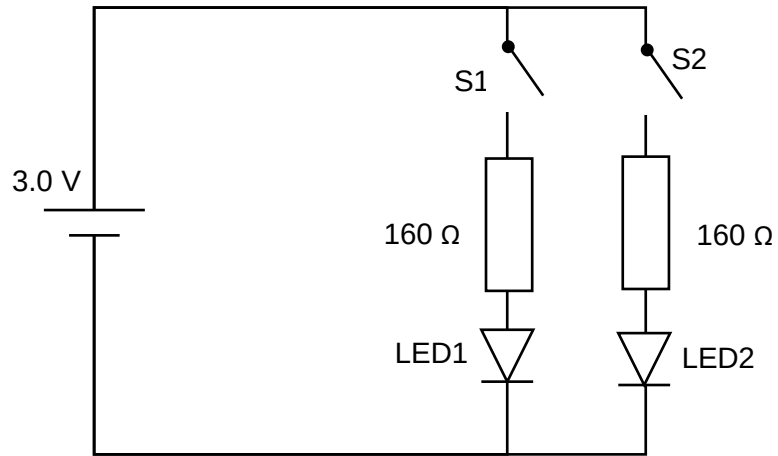
OR

With two LEDs in parallel, effective resistance between the LEDs is lowered.

By Potential Divider Principle, the potential difference across each LED is also lowered.

From the graph, it can be seen the current decreases with voltage, so the power ($P=VI$) dissipated in each LED is lower, so brightness is dimmer.

(c)



B1